

YIHANG CHENG

PhD Student in Computer Science, University of Otago

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RESEARCH INTERESTS

Agentic AI for Healthcare · Large Language Model Agents · Clinical NLP ·
Clinical Decision Support · Agent Evaluation and Harness Design

EDUCATION

University of Otago

PhD in Health Informatics

2024 – Present

Dunedin, New Zealand

- Supervisors: Veronica Liesaputra and Andrew Trotman
- Research focus: **Clinical NLP**, **automated medical coding**, and **Agentic AI for Healthcare**

University of Leeds

*MSc Data Science and Analytics, **Distinction***

2021 – 2022

Leeds, United Kingdom

- Supervisor: Samuel Relton
- Dissertation: Early Parkinson's disease diagnosis using transfer learning based on deep learning

PUBLICATIONS

- [1] **Yihang Cheng**, Veronica Liesaputra, and Andrew Trotman. *CoLa-ICD: A Knowledge-Enhanced Framework for Long-Tail Automated Medical Coding*. **Findings of EMNLP 2026**. [\[Paper\]](#)

RESEARCH EXPERIENCE

CoLa-ICD: Long-Tail Automated Medical Coding

PhD Research, University of Otago

2024 – Present

Dunedin, New Zealand

- Proposed **CoLa-ICD** to improve rare-code prediction in automated medical coding by jointly modeling clinical terminology, inter-code dependencies, and alignment between clinical evidence and ICD labels.
- Developed graph-based label representations, terminology-aware attention, and label-aware contrastive learning; achieved **0.108 Macro-F1** on MIMIC-IV ICD-10 versus 0.069 for GKI-ICD, and **0.204 Micro-F1** on codes with 1–10 training examples versus 0.127.

Sequential Clinical Diagnosis with LLM Agents

PhD Research, University of Otago

2024 – Present

Dunedin, New Zealand

- Built an **agentic clinical reasoning framework** in which LLM agents interact with a simulated patient environment, iteratively selecting questions and tests, maintaining diagnostic state, and updating differential diagnoses from newly acquired evidence.
- Developed trajectory-generation and **reinforcement-learning** components for agent policy optimization, and evaluated agent behavior in terms of diagnostic accuracy, evidence acquisition, interaction efficiency, and consultation cost.

Automated Video Assessment of Parkinsonian Bradykinesia

MSc Dissertation, University of Leeds

2022

Leeds, United Kingdom

- Developed an **end-to-end pipeline** for automated analysis of hundreds of smartphone-recorded finger-tapping videos, combining YOLOv5 hand localization with deep video classification.
- Compared Video Swin Transformer, I3D, TSN, and C3D under same settings; the ImageNet-pretrained Video Swin Transformer achieved the best held-out performance (**87.5% accuracy; 0.938 AUC**).

SELECTED PRESENTATIONS

EMNLP 2026

Poster

2026

Budapest, Hungary

- CoLa-ICD: A Knowledge-Enhanced Framework for Long-Tail Automated Medical Coding

HINZ / AI in Health Pitch Fest

Pitch

2026

Dunedin, New Zealand

- How Can AI Support Doctors in the Clinical Workflow?

TEACHING EXPERIENCE

Tutor, COSC326 Computational Problem Solving 2025
University of Otago *Dunedin, New Zealand*

- Delivered tutorials and programming laboratories for **approximately 70 students**, supporting algorithmic reasoning, computational problem solving, and programming practice.
- Designed assessment questions and programming problems, assessed student submissions, and provided feedback on algorithmic approaches, code correctness, and problem-solving strategies.

ACADEMIC SERVICE AND PROFESSIONAL ACTIVITIES

Postgraduate Representative 2026 – Present
University of Otago *Dunedin, New Zealand*

- Represent the postgraduate student community in School-level discussions, synthesizing student perspectives and communicating priorities, concerns, and feedback to academic staff.
- Contribute to postgraduate engagement and research-community development, including coordinating the **School of Computing postgraduate symposium**, supporting program planning, presenter liaison, and student participation.

Student Volunteer, EMNLP 2026 2026

Member, AI in Health Research Network, New Zealand

AWARDS AND SCHOLARSHIPS

Data Science Scholarship, University of Leeds 2021

First-Class Scholarship, East China Jiaotong University 2020

Second Prize, China University Business Challenge 2019

SELECTED ACADEMIC TRAINING

Oxford Machine Learning Summer School (OxML 2022) 2022

MLx Health & Bio · MLx Finance

- Completed advanced training in machine learning with specialised applications in healthcare, biomedicine, and quantitative finance.
- Health & Bio: Developed and evaluated machine-learning models for **mortality prediction** using MIMIC clinical data; presented the methodology and empirical findings.
- Finance: Applied deep learning to financial factor discovery and quantitative modelling, focusing on predictive signal extraction for systematic trading; presented the methodology and results.

INDUSTRY EXPERIENCE

Data Analyst 2022 – 2024
Sensors Data *Shenzhen, China*

- Utilized Python, SQL, statistical analysis, and machine learning techniques to analyze large-scale product data, focusing on **user segmentation, behavioral modeling**, funnel analysis, retention, and conversion rates.
- Designed and evaluated **A/B tests** while contributing to data governance through metric standardization, data quality validation, and dashboard development. Translated quantitative findings into actionable recommendations for clients and cross-functional stakeholders.

TECHNICAL SKILLS

Programming: Python, SQL, R

ML/NLP: PyTorch, Transformers, SFT, LoRA, reinforcement learning, agent evaluation

Systems: Linux, Slurm/HPC, Git

Languages: Mandarin Chinese (native); English (fluent)